
How Do World Action Models Use Their Futures?

Evidence from Mechanistic Interventions

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Abstract

World action models generate future observations alongside actions, but those futures may guide behavior or merely accompany actions computed elsewhere. Destructive ablations establish dependence, not whether coherent future content directionally controls behavior. We introduce reachable-donor future transplantation: between two continuations from the same restored state, we substitute one policy-generated future while holding observation, instruction, and random draws fixed. The alternative predicts a specific direction of behavioral change. Donor futures nearly completely redirect Cosmos 3 actions and endpoints across 22 states and six RoboLab tasks; Cosmos Policy shows related early-state steering across all ten LIBERO-Long tasks. Cosmos Policy responds primarily to visible robot motion, whereas neither robot-only nor object-only pixels reproduce Cosmos 3’s whole-future effect. Replacing donor-future attention keys and values with self-future counterparts removes 83–88% of Cosmos 3 steering. These results establish content-specific future conditioning and reveal distinct mechanisms across the two policies.

1 Introduction

World action models (WAMs) generate future observations alongside robot actions [8, 14], but joint generation does not imply causal use. Futures might be auxiliary outputs, a generic workspace, an inverse-dynamics input such as prospective robot pose [1, 7], or predicted consequences used to compare actions [2, 3]. Content-specific conditioning alone would not establish outcome comparison. We study direct-policy inference in Cosmos Policy and the diffusion generator in Cosmos3-Nano-Policy-DROID; neither procedure samples and scores multiple candidate actions.

RIFT showed that masking or editing future-position values changes WAM trajectories and success [22]. Such destructive interventions establish dependence, but damage can change an action arbitrarily and therefore does not distinguish generic disruption from content-specific use.

Our test is directional. From the same restored state, two native policy continuations define a recipient and a donor. Both are coherent alternatives generated by the unmodified policy, and their one-chunk endpoints are reachable from that state. We insert the donor future into the recipient computation while retaining the recipient’s current inputs and random draws. The model never receives the donor action; we use it only afterward to define the direction in which we score the recomputed action and its executed simulator endpoint (Figure 1). A generic perturbation need not move either quantity toward that particular donor. Directional movement therefore tests content-specific control, rather than dependence on the future channel alone.

We ask whether coherent futures steer behavior, which content suffices, and which interface carries the effect. Cosmos 3 steering generalizes across 22 states, is not reproduced by isolated robot or object pixels, and is suppressed by future-token K/V replacement. Cosmos Policy effects con-

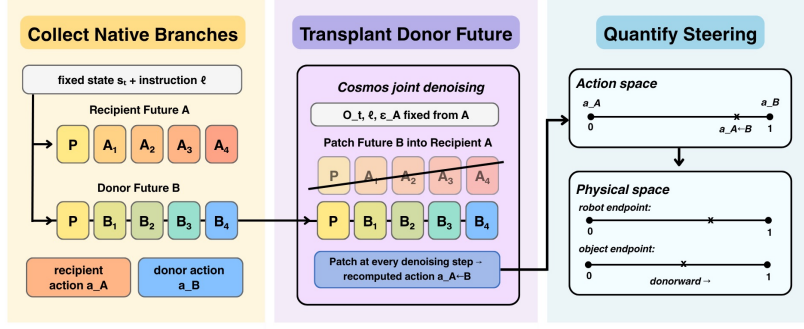


Figure 1: Reachable-donor transplantation. Two native continuations share a restored state; only the donor future is substituted. The withheld donor action defines signed movement from recipient (0) toward donor (1). Self-future K/V replacement tests the pathway.

centrate at early states and are most consistent with camera-visible robot motion. This supports future-dependent action generation, not candidate search or planning.

2 Related Work

Prediction and control. Some visual robot policies generate future frames and infer actions from them; others condition action inference on predicted video representations [1, 7]. Joint video–action models instead learn future observations and controls in one model [6, 8, 10, 14, 20]. Their training objective does not determine whether a particular generated future is used at inference. UVA, for example, can omit video generation at test time [10]. Explicit visual planners answer a different question by sampling alternatives, predicting their outcomes, and comparing them with a goal-dependent score [2, 3].

Recent studies separate predictive training from using predictions at inference. Fast-WAM removes explicit test-time rollout [21]; Faster-WAM retains a cheaper conditioning route [24]; and AGRA finds that plausible foresight need not support reliable action inference [16]. RIFT establishes dependence on organized future-position values [22]; we ask whether a coherent alternative predicts the direction of behavior.

Mechanistic interventions. Activation replacement tests causal influence, and path patching restricts it to specified routes [5, 12, 18]. Results depend on the counterfactual and metric, while routes can interact [17, 23]; we therefore interpret K/V replacement as an active pathway, not a complete circuit.

3 Methods

3.1 Future transplantation and directional projection

Let saved simulator state S admit two unmodified branches A and B , with future representations F_A, F_B , action chunks $\mathbf{a}_A, \mathbf{a}_B$, and endpoints $\mathbf{e}_A, \mathbf{e}_B$ obtained by executing one action chunk. We call A the recipient—the branch whose action is recomputed—and B the donor—the branch supplying the alternative future. To recompute A , we restore the same state and observation, retain its instruction, action noise, future-noise path, solver, and denoising schedule, and replace only F_A with F_B . We repeat the intervention in the reciprocal direction, transplanting F_A into B .

For an action or endpoint $\mathbf{x} \in \{\mathbf{a}, \mathbf{e}\}$, normalized donor projection is

$$\pi_x(\hat{\mathbf{x}}; A, B) = \frac{(\hat{\mathbf{x}} - \mathbf{x}_A)^\top (\mathbf{x}_B - \mathbf{x}_A)}{\|\mathbf{x}_B - \mathbf{x}_A\|_2^2}. \quad (1)$$

The recipient is 0, the donor is 1, and 0.5 is halfway between them along the donor direction; negative values move away and values above 1 overshoot. Because the dot product retains only this signed

direction, a large change perpendicular to the donor receives little projection. The primary steering contrast subtracts the corresponding self-future recomputation:

$$\Delta_x = \pi_x(\hat{\mathbf{x}}_{\text{donor}}; A, B) - \pi_x(\hat{\mathbf{x}}_{\text{self}}; A, B). \quad (2)$$

The self condition reruns the same intervention machinery with the recipient’s own future, so subtraction removes any displacement due merely to recomputation. Pairs are selected using native action and endpoint separation, never intervention outcomes. Every endpoint arm is executed after restoring the exact simulator state.

3.2 Models, populations, and controls

Cosmos Policy. We evaluate the public `Cosmos-Policy-LIBERO-Predict2-2B` checkpoint in direct-policy mode on LIBERO-Long, also called LIBERO-10 [8, 11]. The frozen grid contains 30 states: ten tasks crossed with the first query, a state after three open-loop chunks, and a state after six chunks. Eight native runs determine one reciprocal donor pair per state; three future-noise draws are repeated within direction. Endpoints concatenate goal-relevant simulator coordinates with the official nine-dimensional proprioceptive endpoint.

Cosmos 3. We evaluate the 16B `Cosmos3-Nano-Policy-DROID` checkpoint on six eligible RoboLab tasks [14, 15, 19]. Its diffusion generator has 36 two-way full-attention layers, four denoising calls, 33 raw video frames mapped to nine VAE latent frames, and 32 absolute joint-position actions. The transplant targets unconditioned future-video coordinates at the final guided-sampler boundary. A frozen screen selected 24 state clusters, capped at four per task, using only native success, action and endpoint separation, and replay validity. Twenty-two completed. Two source branches ended before producing full 33-frame videos and were left missing rather than replaced. Reciprocal directions and predicted versus executed future sources are repeated measurements within state.

Controls include exact self-future recomputation, Gaussian targets matched to the donor in norm and recipient distance, a prespecified natural-future control, within-task shuffled futures, and bitwise no-op checks. Gaussian controls test whether an equally large unstructured displacement is sufficient; the natural and shuffled controls compare the paired donor against a prespecified non-donor continuation and mismatched futures. For Cosmos 3, identity layout capture, sampler wrapping, cache record/replay, and self patches reproduce actions bitwise; sentinels confirm that the transplant does not directly write to action coordinates. Executed-future conditions encode videos produced by physically executing reachable donor actions and replay the same environment prefix before intervention. They test whether steering survives a physically realizable continuation rather than appearing only for model-predicted imagery.

3.3 Robot and object interventions

We isolate content in complementary ways. In Cosmos Policy, first-query interventions replace the wrist-camera future, primary-camera future, or future proprioception separately. A second study crosses binary robot and object states in re-rendered videos at 20 held-out states. A decoder checks whether the intended cell remains visible. A natural-pair screen separately searches policy-generated continuations for robot-divergent/object-matched and object-divergent/robot-matched futures; only four states meet its criteria, below the planned minimum of ten.

For Cosmos 3, a separately frozen factor study selected 12 population states round-robin by native object-position separation. Ten completed across all six tasks. Starting from the recipient video, an object-priority disjoint mask copies donor object pixels, donor robot pixels outside the object mask, both, or neither. This 2×2 design tests isolated pixel factors; it does not guarantee an in-distribution video or isolate a latent semantic variable.

3.4 K/V interventions

In attention, keys help determine what an action query reads, while values carry the information returned by that read. In Cosmos Policy, action queries can attend to future keys. We continuously gate future keys at prespecified block 27, with exact all-key recomputation, equal-count random-key and current-key controls, and a block-0 control. This removal changes attention normalization, so

116 it tests whether a late future-to-action edge is active, not the information carried by a token-count-
117 preserving route.

118 Cosmos 3 permits the stronger replacement test. We record future-video keys and values from a
119 self-future run, then substitute them for donor-run future-token K/V at the frozen direct interface.
120 Token count, token order, current-video K/V, text K/V, action K/V, observation, instruction, and
121 noise remain fixed. Layer and pathway choices were frozen on excluded calibration tasks before
122 the 22-state population intervention. We report *K/V replacement loss*: the reduction in donor projec-
123 tion after substituting self-future K/V. This is an operational intervention contrast, not a population
124 natural indirect effect.

125 3.5 Statistical analysis

126 The independent unit is a restored simulator state, not an individual noise draw or transplant direc-
127 tion. We first average reciprocal directions, donors, noise draws, and process repeats within state,
128 then form 95% percentile intervals from 10,000 bootstrap resamples of states [4]. For the common-
129 scale condition estimates in Table 1, we reconstruct each control’s projection within unit from the
130 registered pairwise contrasts before bootstrapping; these are not differences of aggregate means.
131 Task-balanced sensitivity analyses average states within task before resampling tasks. We prespec-
132 ified 0.10 projection units—10% of the native recipient-to-donor displacement—as the smallest ef-
133 fect of interest for the Cosmos Policy confirmatory family and the Cosmos 3 population and factor
134 families [9]. An interval contained within $[-0.10, 0.10]$ supports practical equivalence to zero at
135 that scale. Prespecified primary sign-test p -values use Holm correction.

136 4 Results

137 4.1 Do coherent futures steer behavior?

138 The method succeeds if three things happen: inserting a donor future moves behavior toward that
139 donor, the shift survives simulator execution, and the paired donor outperforms non-donor controls.
140 Table 1 puts the method and every applicable control on one scale, organized by model, future
141 source, and outcome. Figure 2 separates the two statistical questions. Panel a reports donor-minus-
142 self steering, for which positive values indicate movement toward the paired donor and 1 denotes
143 the full native recipient-to-donor displacement. Panel b reports donor-minus-control contrasts, for
144 which positive values mean that the paired donor steers behavior more strongly than the stated
145 control.

146 **Cosmos 3.** Across 22 completed states and all six tasks, a predicted paired-donor future produces
147 action steering of 0.992 ($[0.972, 1.010]$; 22/22 states positive), compared with 0.559 for the natural-
148 future control (Table 1). Executed paired-donor videos produce action steering of 0.983, compared
149 with 0.591 for the natural control and 0.041 for the geometry-matched Gaussian control. Executing
150 the paired-donor-conditioned action yields endpoint steering of 1.003 ($[0.950, 1.050]$; 22/22). The
151 corresponding endpoint projections are 0.514 ($[0.387, 0.642]$) for the natural control and -0.041
152 ($[-0.168, 0.054]$) for the matched Gaussian. In paired secondary analyses, the donor therefore ex-
153 ceeds the natural endpoint control by 0.488 ($[0.348, 0.631]$; 21/22 states positive) and the Gaussian
154 endpoint control by 1.043 ($[0.946, 1.168]$; 22/22). Thus, the result is not explained by recomputa-
155 tion, an equally large unstructured displacement, or merely supplying a different valid continuation.
156 Endpoint projection shows that the donor-directed component survives simulator execution; it does
157 not imply equality of the full endpoint vectors. Task-balanced analyses preserve these conclusions.

158 **Cosmos Policy.** At the first query, the paired donor produces action steering of 0.499 and end-
159 point steering of 0.552, positive in all ten tasks (Table 1). The natural control reaches 0.395 and
160 0.442, respectively, while the shuffled control reaches 0.244 and 0.275 and the matched Gaussian
161 remains near zero. The natural future is the hardest control: the paired donor exceeds it by 0.103
162 ($[0.048, 0.161]$; 8/10) for actions and 0.110 ($[0.054, 0.166]$; 8/10) for endpoints. This rules out the
163 weak explanation that any different tokenizer-consistent future causes the same movement. Figure 5
164 examines how this effect varies across state cohorts and future components.

Table 1: Method and controls on one scale. Each row reports donor-directed projection after inserting the named future: 0 is the self-future result and 1 is the paired donor behavior. Within each model–source–outcome block, bold marks the largest estimate, always the paired donor. Control projections are reconstructed within independent unit before bootstrapping; the two Cosmos 3 endpoint-control rows are secondary analyses.

Model	Source	Outcome	Future inserted	Projection [95% CI]
Cosmos 3	Predicted	Action	Self	0.000 (reference)
Cosmos 3	Predicted	Action	Natural control	0.559 [0.465, 0.651]
Cosmos 3	Predicted	Action	Paired donor	0.992 [0.972, 1.010]
Cosmos 3	Executed	Action	Self	0.000 (reference)
Cosmos 3	Executed	Action	Matched Gaussian	0.041 [0.003, 0.076]
Cosmos 3	Executed	Action	Natural control	0.591 [0.491, 0.687]
Cosmos 3	Executed	Action	Paired donor	0.983 [0.890, 1.064]
Cosmos 3	Executed	Endpoint	Self	0.000 (reference)
Cosmos 3	Executed	Endpoint	Matched Gaussian	−0.041 [−0.168, 0.054]
Cosmos 3	Executed	Endpoint	Natural control	0.514 [0.387, 0.642]
Cosmos 3	Executed	Endpoint	Paired donor	1.003 [0.950, 1.050]
Cosmos Policy	Predicted	Action	Self	0.000 (reference)
Cosmos Policy	Predicted	Action	Matched Gaussian	0.001 [−0.004, 0.006]
Cosmos Policy	Predicted	Action	Natural control	0.395 [0.269, 0.553]
Cosmos Policy	Predicted	Action	Shuffled control	0.244 [0.152, 0.347]
Cosmos Policy	Predicted	Action	Paired donor	0.499 [0.336, 0.679]
Cosmos Policy	Predicted	Endpoint	Self	0.000 (reference)
Cosmos Policy	Predicted	Endpoint	Matched Gaussian	0.003 [−0.003, 0.008]
Cosmos Policy	Predicted	Endpoint	Natural control	0.442 [0.311, 0.598]
Cosmos Policy	Predicted	Endpoint	Shuffled control	0.275 [0.185, 0.375]
Cosmos Policy	Predicted	Endpoint	Paired donor	0.552 [0.387, 0.728]

No screened branch pair in either policy retained a stable success/failure label across continuation seeds. We therefore interpret these results as donor-directed changes in action and one-chunk endpoint, not changes in closed-loop task completion.

4.2 What future content causes steering?

The content that drives steering differs across checkpoints. Cosmos Policy responds mainly to camera-visible robot motion at early states, whereas neither isolated pixel factor recreates Cosmos 3’s whole-future effect.

Cosmos Policy. Wrist-future action steering is 0.435 ([0.284, 0.605]), compared with 0.101 ([0.057, 0.144]) for the primary camera and 0.001 ([−0.004, 0.008]) for future proprioception (Figure 5b). In the 20-state 2×2 study, decoded futures identify the intended cell with 92.5% primary-camera and 98.75% wrist-camera accuracy. The intended object manipulation is therefore visible in the rendered futures, yet the object-state main effect is 0.00015 ([−0.00012, 0.00049]) for action and 0.00018 ([−0.00040, 0.00102]) for the executed goal-state endpoint. The available analysis does not report the robot-state main effect. Four exploratory natural pairs provide a complementary comparison: robot-motion futures yield action steering of 0.291 ([0.060, 0.521]) and robot-endpoint steering of 0.315 ([0.059, 0.571]), whereas the object-only effects are 0.021 for both action and goal endpoint, with intervals crossing zero. At these sampled early states, the pattern is most consistent with an inverse-dynamics-like role for camera-visible robot motion. It does not show that Cosmos Policy generally ignores object consequences.

Cosmos 3. Across ten factor-study states and all six tasks, every isolated-factor interval lies inside the prespecified [−0.10, 0.10] equivalence region (Table 2; Figure 3). Panels a and b show all four cells of the 2×2 hybrid intervention: swapping robot pixels, object pixels, or both changes neither the action nor the executed endpoint appreciably. Panel c places the factor effects beside the whole-future intervention on the same axis. Whole-future steering is approximately one unit, whereas robot-pixel effects are 0.0022 for actions and 0.0102 for endpoints, and object-pixel effects are 0.0061 and

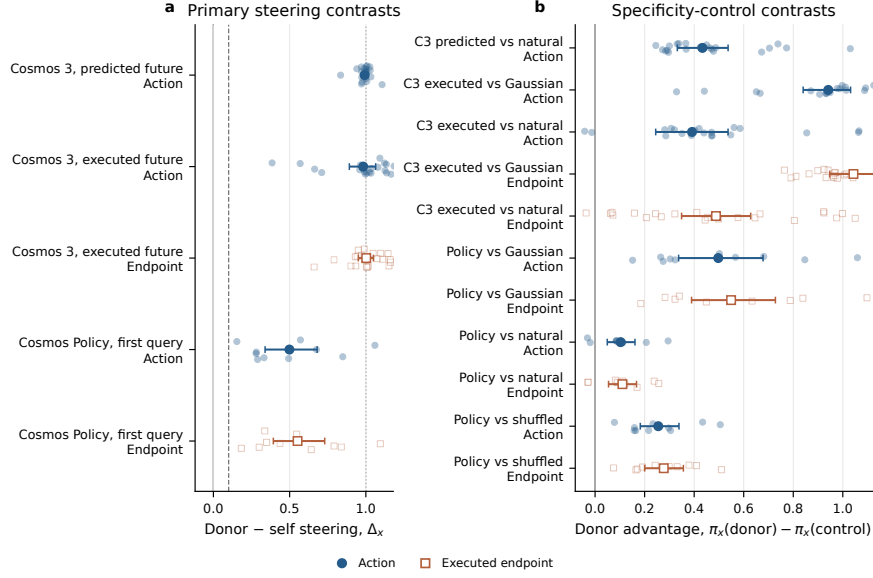


Figure 2: Directional steering and specificity controls. (a) Donor-minus-self steering. Values near 1 mean that the intervention reproduces the donor-directed component of the native action or endpoint displacement; the dashed line marks the prespecified 0.10 smallest effect of interest. (b) Paired donor minus the stated control. Positive values mean that the paired donor has greater directional influence than an unstructured, natural, or shuffled alternative. Small symbols are independent states for Cosmos 3 and tasks for Cosmos Policy; large symbols and whiskers are means and 95% independent-unit bootstrap intervals. Blue circles denote actions and open orange squares denote executed endpoints.

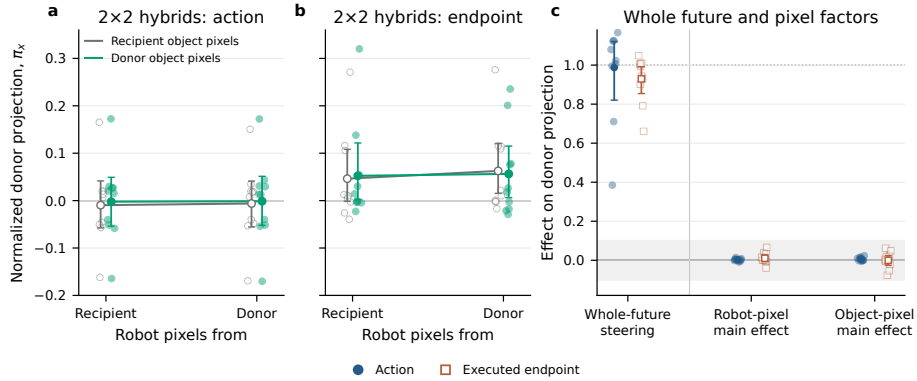


Figure 3: Cosmos 3 robot/object pixel factorization. Panels (a) and (b) show the four hybrid-video cells for action and executed endpoint projection. The horizontal axis indicates whether robot pixels come from the recipient or donor; gray and green traces indicate recipient and donor object pixels. Panel (c) compares the resulting robot- and object-pixel main effects with transplantation of the complete donor future. Blue circles denote actions and open orange squares denote executed endpoints; small symbols are states and large symbols with whiskers are means and 95% state-bootstrap intervals. The gray band is the prespecified $[-0.10, 0.10]$ equivalence region. Isolated robot or object pixels have near-zero effects even though the complete future strongly steers both outcomes.

190 -0.0001 . Neither isolated factor therefore recreates a meaningful fraction of the whole-future effect.
 191 The useful signal may be distributed across the image, require robot-object coherence, depend on
 192 camera or temporal structure, or interact nonlinearly with masking and tokenization. Hybrid videos

may also be off distribution. The experiment establishes insufficiency of each isolated pixel factor, not irrelevance of robot or object information and not joint necessity.

Table 2: Cosmos 3 isolated-content interventions across 10 states from six tasks. Entries are mean donor-directed projections with 95% state-bootstrap intervals; 0 denotes the recipient and 1 the paired donor. Robot and object masks are disjoint. The prespecified practical-equivalence region is $[-0.10, 0.10]$.

Donor pixels inserted	Action projection [95% CI]	Endpoint projection [95% CI]
Robot only	0.0022 $[-0.0016, 0.0060]$	0.0102 $[-0.0067, 0.0275]$
Object only	0.0061 $[0.0009, 0.0119]$	-0.0001 $[-0.0254, 0.0239]$

4.3 Which internal pathway carries the effect?

Future-token attention provides an active route from generated futures to action. Population-level K/V replacement in Cosmos 3 gives the stronger of the two pathway tests.

Cosmos Policy. Block-27 future-key removal produces a monotonic action-disruption curve: normalized L_2 change is 0.022, 0.035, 0.054, and 0.068 at gates 0.25, 0.50, 0.75, and 1.00. All-key recomputation is bitwise exact. Full removal is 0.016 more disruptive than equal-count random-key removal and 0.048 more disruptive than block-0 future-key removal; equal-count current-key removal is 0.024 larger. Executed endpoints support the same bounded conclusion, although state counts and intervals are not available for this analysis. The late future-to-action edge is active, but it is neither the only nor the largest route through the block.

Cosmos 3. The interface was selected on excluded calibration tasks and frozen before population testing. Replacing donor-run future-token K/V with self-future K/V removes 0.877 of the 0.992 predicted-action shift, 0.814 of the 0.983 executed-action shift, and 0.851 of the 1.003 endpoint shift (Table 3; Figure 4). These losses correspond to 88%, 83%, and 85% of the respective unpatched point estimates and are positive in 22/22 states. The remaining mean steering after replacement is 0.115, 0.169, and 0.152, respectively (Figure 4a–c). Thus, this interface carries most, but not all, of the donor-directed effect. Task-balanced loss estimates are 0.877, 0.812, and 0.853; all prespecified primary sign tests remain significant after Holm correction. Because token count and attention positions remain fixed, donor-specific information travels through this future-token K/V interface. The experiment does not identify every head or feature, exclude interacting unpatched routes, or establish a complete circuit.

Table 3: Cosmos 3 future-token pathway intervention across 22 states from six tasks. Donor steering is the unpatched donor-directed projection. Replacement loss is the reduction in that projection after replacing donor-future K/V with self-future K/V; intervals are 95% state-bootstrap intervals. Relative reduction is the replacement-loss point estimate divided by unpatched donor steering.

Outcome	Donor steering	Replacement loss [95% CI]	Relative reduction
Predicted action	0.992	0.877 $[0.842, 0.910]$	88%
Executed action	0.983	0.814 $[0.723, 0.900]$	83%
Executed endpoint	1.003	0.851 $[0.766, 0.922]$	85%

4.4 Where is the Cosmos Policy effect concentrated?

Figure 5 summarizes three scope checks for Cosmos Policy. Panel a shows a strong first-query effect, followed by near-zero estimates in the separately sampled states after three and six open-loop chunks: action and endpoint steering are 0.031 and 0.004 after three chunks, and 0.027 and 0.025 after six. This demonstrates state dependence associated with query time, not causal decay within a fixed state. Panel b shows that the early action effect is concentrated in the wrist-camera future (0.435), with a smaller primary-camera effect (0.101) and no measurable proprioceptive effect (0.001). Panel c asks how much more strongly the paired donor steers than each control. Its roughly 0.5 donor-minus-self and donor-minus-Gaussian values reflect the primary effect and the ineffectiveness of matched

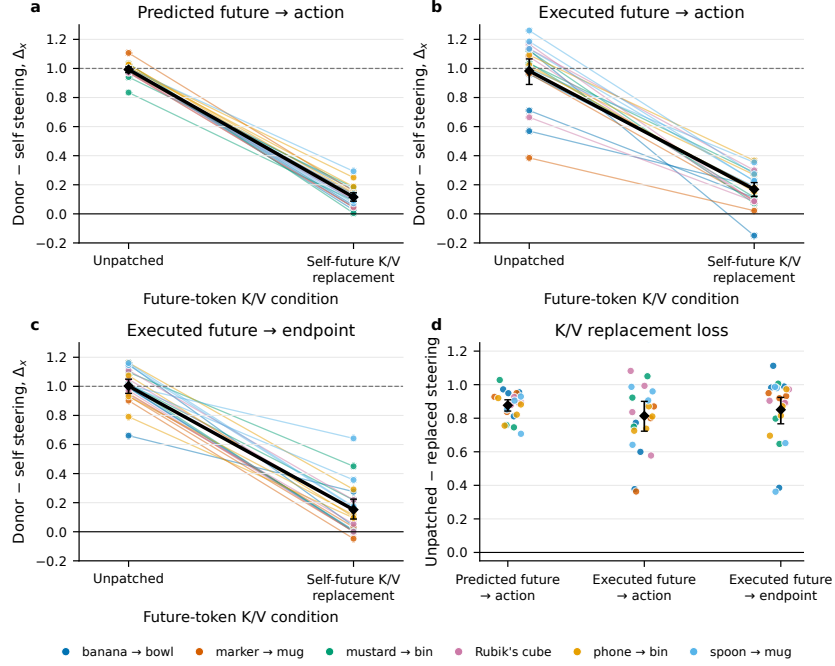


Figure 4: Cosmos 3 future-token K/V replacement. Panels (a)–(c) connect each state’s unpatched donor steering to its value after donor-future keys and values are replaced by self-future keys and values. The black diamonds and whiskers are means and 95% state-bootstrap intervals; colored points identify the six RoboLab tasks. Replacement reduces predicted-action, executed-action, and executed-endpoint steering in all 22 states, leaving mean residual steering of 0.115, 0.169, and 0.152. Panel (d) reports the same intervention as replacement loss (unpatched minus replaced); positive values indicate that the pathway carries donor-specific influence.

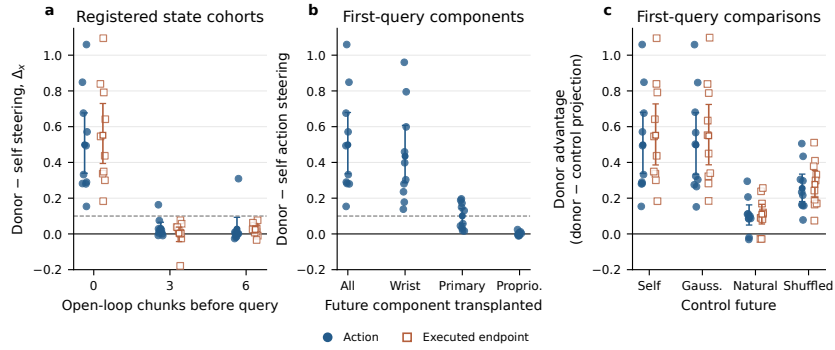


Figure 5: Scope of the Cosmos Policy effect. (a) Paired-donor steering for the first query and for distinct states sampled after three and six open-loop chunks. (b) First-query action steering when all future inputs, only the wrist-camera future, only the primary-camera future, or only future proprioception is transplanted. (c) First-query donor advantage over self, matched-Gaussian, natural-future, and shuffled-future controls. Blue circles denote actions and open orange squares denote executed endpoints; small symbols are tasks and large symbols with whiskers are means and 95% task-bootstrap intervals. The dashed line marks the prespecified 0.10 smallest effect of interest.

noise; the smaller natural-control advantage (0.103 for action and 0.110 for endpoint, each positive in 8/10 tasks) is the more stringent content-specific comparison. The shuffled-control advantage is intermediate. Together, these panels localize the Cosmos Policy result to early visual futures while retaining an advantage over other valid futures.

229 5 Discussion and Limitations

230 Across both WAM generations, transplanting a coherent future predicts which alternative action
231 the model will approach. The associated simulator endpoint moves in the same direction. RIFT
232 established that behavior depends on an organized future interface; our result adds content-specific
233 sufficiency by showing that a policy-native alternative predicts the signed direction of the change.
234 The two checkpoints, however, do not appear to rely on the same content.

235 Cosmos Policy’s early-state pattern resembles an inverse-dynamics interface conditioned on visible
236 robot motion. That account does not transfer cleanly to Cosmos 3: strong whole-future steering
237 coexists with negligible robot-only and object-only pixel effects. Usable content may instead depend
238 on architecture, surrounding context, or distributed feature interactions. The pathway experiments
239 locate where some of this influence travels. Cosmos Policy identifies an active late attention edge,
240 while Cosmos 3 shows across 22 states that future-token K/V carries most of the donor-directed
241 shift.

242 This evidence is narrower than planning. A planning claim would require evidence that the policy
243 generates alternatives, predicts their outcomes, compares them using a goal, cost, or value, and se-
244 lects accordingly. We do not test that sequence, and no screened pair retained a stable success/failure
245 label across continuation seeds. Here, “future” and “imagined” describe generated representations,
246 not deliberation. Our claim is that these direct policies condition action on coherent futures, not that
247 WAMs generally plan over task consequences.

248 **Limitations.** All interventions are in simulation and cover two checkpoints, six RoboLab tasks,
249 and LIBERO-Long. Cosmos Policy’s strongest result uses one first-query state per task, while later
250 estimates use different physical states and therefore cannot isolate elapsed time. Its natural content
251 analysis contains only four states, and rendered hybrids may be outside the policy’s native future
252 distribution. Cosmos 3 completed 22 of 24 frozen population states and 10 of 12 factor states; the
253 two population omissions resulted from short source videos and were not replaced. Pixel masks
254 may miss distributed semantic features or induce nonlinear changes, so negligible isolated effects
255 do not establish joint necessity. Key removal changes attention normalization, and K/V replacement
256 can interact with unpatched routes. We do not demonstrate task-success mediation or recover a full
257 circuit.

258 **Broader impact.** Mechanistic tests can help distinguish plausible-looking predictions from repre-
259 sentations that actually control behavior. The same interfaces could also redirect robot actions or
260 create false confidence if validated on narrow state distributions. Use beyond simulation would re-
261 quire broader coverage, physical safety constraints, and monitoring outside the intervention metric.

262 6 Conclusion

263 Reachable-donor transplantation turns the broad question of whether a world action model “uses”
264 its future into a directional causal test. In Cosmos 3, predicted and physically executed donor fu-
265 tures produce nearly complete donor-directed action steering across 22 states and six tasks, and the
266 resulting one-chunk endpoints move in the same direction. Replacing donor-future keys and values
267 with self-future counterparts removes 83–88% of this steering in every state. Cosmos Policy shows
268 a smaller but related early-state effect that survives execution and exceeds natural, shuffled, and
269 geometry-matched controls.

270 These experiments establish content-specific future conditioning in two direct policies, not planning
271 over predicted outcomes. Cosmos Policy’s pattern is most consistent with camera-visible robot
272 motion, whereas Cosmos 3’s whole-future effect is not reproduced by either isolated robot or object
273 pixels. Determining whether these mechanisms support consequence-sensitive choice will require
274 donors with stable differences in task success or value, longer-horizon evaluation, and finer-grained
275 interventions on the remaining pathways.

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A Reproducibility and Protocol Details

A.1 Experimental settings and frozen designs

The Cosmos Policy experiments use the public Cosmos-Policy-LIBERO-Predict2-2B checkpoint, LIBERO-Long, PyTorch 2.7 with CUDA 12.8, and two NVIDIA H100 80GB GPUs. The frozen design contains ten tasks, three state indices, eight native policy runs per state, one reciprocal donor pair per state, and three future-noise draws per direction. The observation prefix, instruction, recipient action noise, future-noise schedule, tokenizer settings, and denoising schedule are held fixed within an intervention comparison.

The Cosmos 3 experiments use the public 16B Cosmos3-Nano-Policy-DROID checkpoint with the released Cosmos, Cosmos Framework, and RoboLab evaluation stack. The frozen population screen selected 24 state clusters across six eligible tasks, capped at four clusters per task, using only native success, donor separation, and replay-validity information. Twenty-two clusters completed. The separate robot/object factor study selected 12 clusters by native object-position separation and completed its prespecified minimum of ten across all six tasks.

381 A.2 State restoration, execution, and missingness

382 Each endpoint comparison begins from an identical saved simulator state. Evaluation reconstructs
383 the environment, restores that state, replays the observation prefix, verifies the restored observation,
384 and then executes one recomputed action chunk. Recipient and donor endpoints are produced by the
385 same restoration and execution procedure. This design makes the endpoint projection a comparison
386 between reachable continuations of one physical state rather than between unrelated trajectories.

387 Two frozen Cosmos 3 population units were unavailable because their registered source branches
388 ended after 21 and 29 actions, before producing the required 32-action, 33-frame intervention source.
389 Neither unit was replaced, and no intervention outcome entered the decision to retain or omit a state.
390 The completed population therefore exceeds the prespecified minimum of 20 states across at least
391 five tasks. The factor study likewise meets its minimum of ten states across at least five tasks.

392 A.3 Intervention validity and controls

393 The self-future condition passes the recipient’s own future through the complete intervention ma-
394 chinery. For Cosmos 3, identity token-layout capture, sampler wrapping, cache record/replay, and
395 self-patching reproduce baseline actions bitwise. Sentinels at the sampler boundary confirm that
396 future transplantation does not write directly into action coordinates. The future-token pathway
397 experiment preserves token count and order while leaving current-video, language, action, and all
398 non-targeted key/value states unchanged.

399 The geometry-matched Gaussian control has the same norm and recipient distance as the donor
400 displacement but no coherent content. Natural controls are valid unpaired continuations selected
401 without intervention outcomes, and shuffled controls use another future from the same task. These
402 comparisons distinguish exact recomputation, perturbation magnitude, generic alternative-future ef-
403 fects, and pair-specific content. Executed-future conditions add a separate realizability check by
404 encoding frames generated when a native donor action is physically replayed in the simulator.

405 A.4 Statistical conventions

406 The independent unit is the restored simulator state. Reciprocal directions, donor choices, noise
407 draws, and repeated processes are averaged within state before any population summary. Reported
408 95% intervals use 10,000 percentile-bootstrap resamples of states; task-balanced checks first average
409 within task and then resample tasks. The 0.10 smallest effect of interest was fixed before the Cosmos
410 Policy confirmatory family and the Cosmos 3 population and factor families were evaluated. Sign
411 tests use state-level directions and Holm correction for each prespecified family.

412 B Additional Experiments and Diagnostics

413 B.1 Condition-level Cosmos 3 projections

414 Figure 6 shows the condition estimates before they are converted into donor-minus-control contrasts.
415 Predicted self-future recomputation stays at zero, natural futures produce partial donor-directed
416 movement, and paired donors cluster near one. For executed futures, the matched Gaussian re-
417 mains near zero while both action and endpoint effects preserve the same ordering. The state-level
418 display also makes clear that the endpoint conclusion is not produced by a single task or a single
419 extreme unit.

420 B.2 Displacement-magnitude diagnostic

421 Directional projection could be misleading if a large normalized estimate arose from an extremely
422 small native denominator or from a negligible transplanted displacement. Figure 7 therefore com-
423 pares unnormalized recipient-to-donor separation with the magnitude induced by transplantation.
424 Predicted-action and executed-endpoint displacements lie close to the identity line across the ob-
425 served range; executed-action displacement is more variable but remains on the scale of the native
426 difference. This diagnostic supports the interpretation that the intervention recreates a substantial
427 behavioral displacement, not merely a tiny change that happens to align with the donor axis.

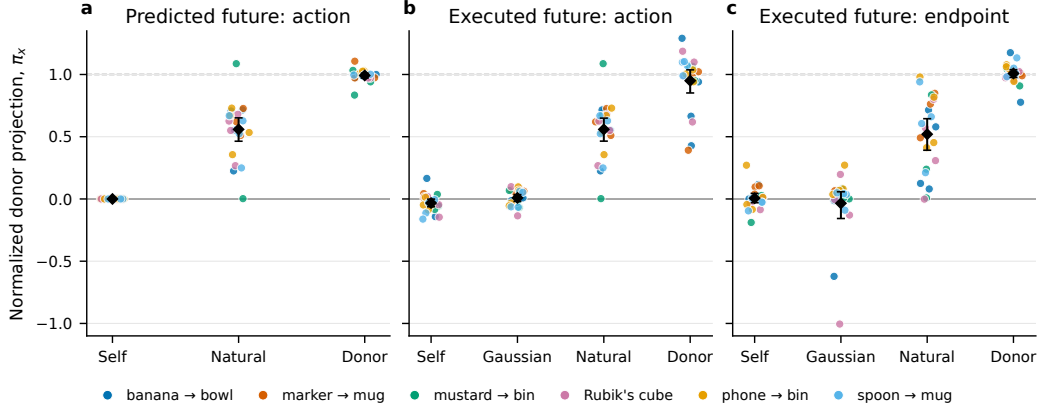


Figure 6: Cosmos 3 condition-level projections across 22 restored states. Panels show predicted-future action, executed-future action, and executed-future endpoint results. Colored points are states grouped by task; black diamonds and whiskers are means and 95% state-bootstrap intervals. Zero denotes the recipient and one the paired donor. Natural controls retain partial donor-aligned content, whereas matched Gaussian targets remain near the recipient and paired donors lie near the native donor.

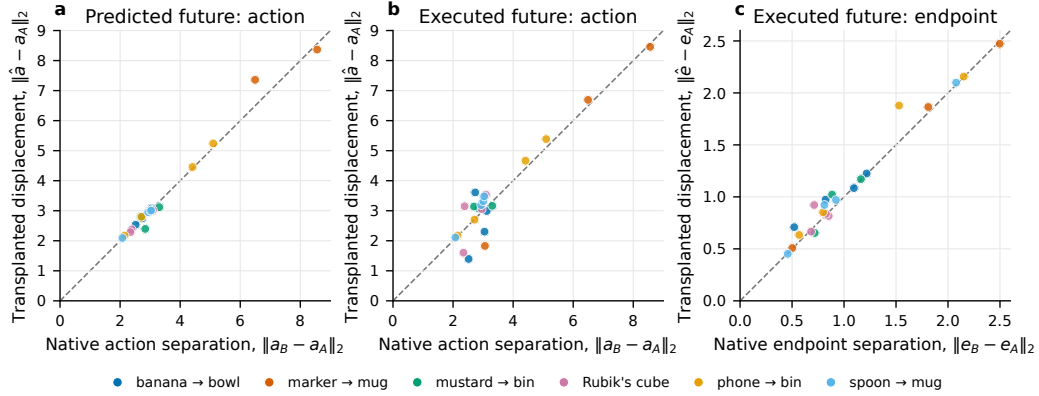


Figure 7: Cosmos 3 magnitude checks. Native recipient-to-donor separation is plotted against the unnormalized displacement caused by donor-future transplantation for predicted actions, executed-future actions, and executed endpoints. The dashed identity line denotes equality of magnitudes; colors identify tasks. These plots complement normalized projection by showing the absolute scale of the induced change.

428 B.3 Cosmos Policy timing and cross-domain checks

429 Across all 30 registered Cosmos Policy states, mean donor-minus-self steering is 0.186
 430 ([0.095, 0.290]) for actions and 0.194 ([0.091, 0.305]) for endpoints. The first-query, after-three-
 431 chunk, and after-six-chunk action estimates are 0.499, 0.031, and 0.027; the corresponding endpoint
 432 estimates are 0.552, 0.004, and 0.025. Because the three indices represent different physical states
 433 that may be nested within task trajectories, the first-query analysis is the cleaner confirmatory result
 434 and the differences should not be read as causal temporal decay.

435 The six-state RoboCasa replication is exploratory and is not pooled with LIBERO [13]. Donor-
 436 minus-self action steering is 0.0144 ([0.0034, 0.0276]), but its donor-minus-Gaussian contrast is only
 437 0.0025 and has an interval crossing zero. Endpoint steering is 0.0287 ([0.0101, 0.0492]), positive
 438 in 6/6 states; donor-minus-Gaussian endpoint steering is 0.0187 ([0.0063, 0.0329]), also positive

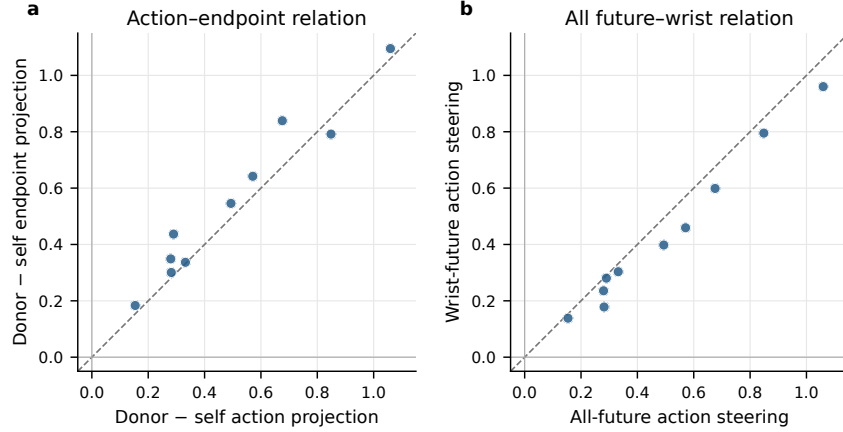


Figure 8: Cosmos Policy task-level relationships at the first query. (a) Donor-minus-self action steering versus executed-endpoint steering. (b) All-future action steering versus wrist-future action steering. Each point is one LIBERO-Long task and the dashed line is identity.

in 6/6. The replication therefore supports a small executed-state effect while leaving action-space specificity uncertain.

B.4 Within-task relationships in Cosmos Policy

Figure 8 provides two task-level diagnostics at the first query. Action and executed-endpoint steering generally rise together, supporting the use of simulator execution as a behavioral continuation of the action result. Wrist-future steering also tracks all-future steering across tasks, consistent with the aggregate modality decomposition while showing that the wrist view does not reproduce the entire effect uniformly.

B.5 Additional content and pathway checks

The Cosmos Policy rendered 2×2 study supplies a manipulation check for the object-content null. Decoded futures identify the intended hybrid cell with 92.5% accuracy in the primary camera and 98.75% in the wrist camera, while executed endpoints identify the four cells at the 25% chance rate. Thus, the future representation visibly contains the imposed distinction even though the pre-registered object main effect on action and goal-state endpoint is approximately zero. The complementary natural-pair search found only four states with both an object-divergent/robot-matched and a robot-divergent/object-matched donor at a common recipient; these exploratory data favor the robot-motion donor, but the available population is below the planned minimum of ten.

The block-27 Cosmos Policy key-removal analysis includes two additional checks beyond the action-space dose curve reported in the main text. Full future-key removal changes the executed endpoint by an absolute mean projection of 0.281 ([0.164, 0.430]; 10/10 first-query states) and exceeds equal-count random-key removal by 0.169 ([0.058, 0.322]; 9/10). Its difference from equal-count current-key removal is -0.035 with an interval crossing zero. The endpoint analysis therefore supports an active future route at the first query while reinforcing the conclusion that this route is not uniquely dominant over current-state information.

For Cosmos 3, task-balanced pathway-loss estimates are 0.877 for predicted actions, 0.812 for executed-future actions, and 0.853 for executed endpoints, closely matching the state-weighted estimates. The remaining steering after self-future K/V replacement is 0.115, 0.169, and 0.152, respectively. This residual motivates finer-grained tests of parallel attention heads, non-attention routes, and interactions between patched and unpatched components rather than a claim that the selected interface is the complete mechanism.